Beyond Matching Objective Values: Exact Decision-Set Audits of Language-Generated Linear Programs

llm4orTT Research Project

Abstract

A language-generated linear program can return the expected objective value while allowing the wrong decisions. We study a decision contract comparing the feasible region and attained optimal face under a fixed variable interpretation. Exact checks preserve a frozen evaluation of 288 responses on 60 held-out problems, where witness repair trails abstract feedback and always-pick-B. Follow-up sampling finds errors in 13/83 fully agreeing six-draw cases. A three-arm feedback ladder on 80 selected problems succeeds on 19, 20, and 23 cases; later rounds add no successes. A metamorphic evaluation yields a reference-conditional false-alarm fraction of 59/287 and triggers its preregistered discard rule. Published references cover 150 original annotation targets; independent human labels were not collected. The results distinguish exact mathematical comparison from reference correctness and identify failures of agreement, richer feedback, and transformation alarms.

Introduction

Natural-language optimization systems translate a verbal decision problem into variables, an objective, and constraints. The NL4Opt benchmark makes this translation task explicit (Ramamonjison et al. 2022). Checking a single optimal value leaves much of the decision problem unexamined: two models can attain the same value at the same point while disagreeing elsewhere.

Our contract checks the whole feasible region and the entire set of attained optima. It separates checking permitted decisions from checking a single returned number. The contribution is an executable audit procedure, a preserved negative repair result, and a reproducible sensitivity analysis of reference readings. The frozen replay is offline. Separately registered follow-ups add sampling, stratified feedback, and metamorphic tests without altering the frozen responses or scores.

Contract and Exact Verification

Let a formulation be $M = (A, b, c, s)$ with a fixed, ordered variable catalog and continuous nonnegative variables. Rewrite its objective as a maximization with vector $\bar{c} = c$ for maximization and $\bar{c} = -c$ for minimization. Define

$$P(M) = \{x \in \mathbb{R}^d : Ax \le b,\ x \ge 0\}.$$

For a feasible model with a finite attained optimum v, define

$$O(M) = \{x \in P(M) : \bar{c}^\top x = v\}.$$

We set $O(M) = \varnothing$ when the model is infeasible or its objective is unbounded. A candidate passes the current-instance contract when both $P(M) = P(G)$ and $O(M) = O(G)$ for a reference G. This contract preserves the attained optimal decisions; it does not require identical ordering of every nonoptimal decision. The original evaluator records that stronger ordering property separately.

For each row $a^\top x \le \beta$ of a target model, inclusion is equivalent to $\sup_{x \in P(M)} a^\top x \le \beta$. Checking all rows in both directions establishes equality. Nonnegativity is included among the rows, and objective equality adds two opposite inequalities when comparing optimal faces. An empty source is contained in every target; an empty target requires an empty source.

Numerical Proposals, Rational Certificates

The new implementation accepts dimensions from one through six. HiGHS supplies numerical solution proposals (Huangfu and Hall 2018). Rational reconstruction checks every primal inequality, dual nonnegativity, stationarity, complementary slackness, and equality of primal and dual objectives using exact fractions. For maximization with all domain constraints represented in $Ax \le b$, a certificate consists of x, y satisfying

$$Ax \le b, \quad y \ge 0, \quad A^\top y = q, \quad q^\top x = b^\top y.$$

Infeasibility uses a Farkas certificate $y \ge 0$, $A^\top y = 0$, and $b^\top y < 0$. Unbounded support requires a feasible point and a recession ray with positive objective gain. A separating point is checked exactly before it is returned.

Reconstruction is bounded by explicit search and arithmetic limits. A failed certificate raises an unresolved result; it is never converted to equivalence. The original two-dimensional reference remains unchanged. Frozen per-case feasible-region and optimal-face comparisons are reproduced by the new implementation; higher dimensions are exercised by analytic regression cases.

Frozen Design and Offline Replay

The frozen cohort contains 60 held-out problems, with development problems retained separately. Each problem supplies two initial candidates, A and B. The abstract repair

arm receives the pair and exact relation diagnostics. The witness repair arm additionally receives concrete separating points when available. Prompt and response files, the call ledger, source snapshots, and the sealed evidence archive are checked before replay.

Table 1 reports all four arms plus selection baselines. Pick-A and pick-B deterministically reuse the corresponding candidate. Oracle-select chooses a correct candidate whenever either candidate passes the reference contract. This is an upper bound for selecting from that pair, rather than an upper bound for generating a new formulation.

The frozen paired comparison of witness versus abstract feedback contains 1 gain and 2 losses, with exact paired test $p = 1.00$. The witness arm remains below both abstract feedback and pick-B in the frozen counts. Marginal success intervals in the tables are equal-tail Clopper–Pearson intervals (SciPy Developers 2026); they describe sampling uncertainty conditional on the reference labels. They are not paired treatment-effect intervals.

Admissible-Reading Sensitivity

A separate review records generated percentage errors, missing terms in official annotations, and ambiguous multiplicative language. Each structured record preserves the quoted span, official constraints, corrected constraints, optional alternative constraints, and exactly checked points. A feasible witness for the corrected reading is stored separately from the original point that separates two formulations.

The sensitivity metric accepts a candidate if it matches at least one admissible reading in both feasible region and optimal face. For a known annotation error, the erroneous official formulation is replaced. For genuinely ambiguous wording, both supported readings are retained. Generated errors alone do not alter a correct reference.

Table 2 retains the frozen scores alongside two requested sensitivity policies. The additional annotation corrections already belong to the seven-case set, so deduplication produces identical policies. The case selection was informed by observed outputs and reviewed by one reader. These results measure sensitivity to specified reading choices. Subsequent reference reuse imports published solutions and checks mathematical validity; it supplies no additional independent reading labels.

Three Observed Failure Modes

Shared errors hide behind agreement. Among 13 pairs in which both candidates fail the frozen reference, 7 pairs agree in feasible region and optimal face. A comparison of A with B produces no separating point for such agreement. Inspecting language anchors or a separately justified reference is necessary to distinguish shared mistakes from acceptable alternative readings.

Reference readings change measured success. A missing annotation term and an ambiguous phrase affect the score even when a geometric comparison is exact. The sensitivity analysis changes all four success counts while B retains the highest count. Mathematical verification establishes consequences of a reading; deciding which reading is supported still requires linguistic evidence.

Concrete witnesses reach a minority of cases. Only 12 of the 60 held-out problems expose an exact witness to repair. The remaining problems provide evidence about the complete prompt pipeline, while witness-specific behavior must be examined on the exposed subset. Within the frozen overall comparison, the observed losses exceed the gains.

Reference Reuse and Follow-up Design

The original annotation plan was amended before the follow-up evaluations to reuse same-version published NL4Opt references. The original 60 IDs and expanded 150 IDs are preserved, with nested cohorts, source hashes, record locations, exact parsing, feasibility, and bounded-optimum checks. Known corrections remain in the separate sensitivity branch. Blank human packages remain archived; no human reading-status labels were collected. Krippendorff alpha is inapplicable to a single-source reference import. One auxiliary model consultation reviewed the protocol without producing per-case annotation labels.

The follow-ups use DeepSeek flash at temperature zero, with continuous nonnegative variables. Each paid stage records its sample size, arms, switching rules, queue hash, and cost estimate before requests. Failures and invalid responses are retained without retries. Selected cohorts include public training and development problems; their scores describe these selected cohorts rather than an independent benchmark test split.

Repeated Sampling (T7)

Six identical-prompt draws per problem are clustered by feasible-region and optimal-face equivalence; the modal class supplies the selected answer. The cohort is deterministically selected from eligible representatives before new outcomes. Temperature zero still permits different observed responses. Table 3 reports modal success and conditional error, retaining incomplete cases separately. Mean modal share is 92.67%, and mean within-problem exact-text repetition is 75.57%; problem bootstrap intervals are in the numerical source. Agreement remains compatible with a shared incorrect formulation.

Feedback Granularity (T8)

Screening uses the original two distinct A/B prompts; T7's identical-prompt draws do not define these strata. Same-version train/dev data supplement local cases. The eligible pool contains 988 problems. Predeclared hash ordering selects 60 divergent pairs and 20 agreeing-but-wrong pairs. Source counts are 37 train, 5 development, and 38 test problems.

Each case starts from A, keeps B fixed, and receives verdict-only, point, or point-plus-anchor feedback for at most three rounds. Anchors are verbatim source spans retrieved heuristically using official constraint annotations; missing anchors remain missing. This assistance is part of the experimental condition. The official formula and correctness label are not supplied in prompts. A case-arm stops after its

Method	Correct/n	Rate (%)	95% CI (%)
pick-B^{R1}	46/60^{R1}	76.7^{R1}	64.0–86.6^{R1}
pick-A^{R1}	42/60^{R1}	70.0^{R1}	56.8–81.2^{R1}
A^{R2}	42/60^{R2}	70.0^{R2}	56.8–81.2^{R2}
B^{R2}	46/60^{R2}	76.7^{R2}	64.0–86.6^{R2}
abstract^{R2}	44/60^{R2}	73.3^{R2}	60.3–83.9^{R2}
witness^{R2}	43/60^{R2}	71.7^{R2}	58.6–82.5^{R2}
oracle-select^{R1}	47/60^{R1}	78.3^{R1}	65.8–87.9^{R1}

Table 1: Frozen-reference results. Pick-A and pick-B reuse the corresponding candidate; oracle-select uses the reference to choose an existing candidate. Every data cell carries its source-directory tag: R1 = `../baselines.json`; R2 = `../rescore.json`. Both paths are relative to this paper's experiment directory. Exact field paths are in `../tables.json`. Intervals are marginal Clopper–Pearson intervals.

Reference policy	Arm	Correct/n	Rate (%)	95% CI (%)	Δk	Rank
Frozen^{R2}	A^{R2}	42/60^{R2}	70.0^{R2}	56.8–81.2^{R2}	+0^{R2}	4^{R2}
Frozen^{R2}	B^{R2}	46/60^{R2}	76.7^{R2}	64.0–86.6^{R2}	+0^{R2}	1^{R2}
Frozen^{R2}	abstract^{R2}	44/60^{R2}	73.3^{R2}	60.3–83.9^{R2}	+0^{R2}	2^{R2}
Frozen^{R2}	witness^{R2}	43/60^{R2}	71.7^{R2}	58.6–82.5^{R2}	+0^{R2}	3^{R2}
Seven-case^{R2}	A^{R2}	46/60^{R2}	76.7^{R2}	64.0–86.6^{R2}	+4^{R2}	4^{R2}
Seven-case^{R2}	B^{R2}	50/60^{R2}	83.3^{R2}	71.5–91.7^{R2}	+4^{R2}	1^{R2}
Seven-case^{R2}	abstract^{R2}	47/60^{R2}	78.3^{R2}	65.8–87.9^{R2}	+3^{R2}	2^{R2}
Seven-case^{R2}	witness^{R2}	47/60^{R2}	78.3^{R2}	65.8–87.9^{R2}	+4^{R2}	2^{R2}
Seven + all annotation^{R2}	A^{R2}	46/60^{R2}	76.7^{R2}	64.0–86.6^{R2}	+4^{R2}	4^{R2}
Seven + all annotation^{R2}	B^{R2}	50/60^{R2}	83.3^{R2}	71.5–91.7^{R2}	+4^{R2}	1^{R2}
Seven + all annotation^{R2}	abstract^{R2}	47/60^{R2}	78.3^{R2}	65.8–87.9^{R2}	+3^{R2}	2^{R2}
Seven + all annotation^{R2}	witness^{R2}	47/60^{R2}	78.3^{R2}	65.8–87.9^{R2}	+4^{R2}	2^{R2}

Table 2: Post-hoc admissible-reading sensitivity. The annotation corrections are a subset of the seven cases, so the two sensitivity policies coincide. Delta is the change from the frozen count; ranks use competition ranking. Every data cell carries its source-directory tag: R1 = `../baselines.json`; R2 = `../rescore.json`. Both paths are relative to this paper's experiment directory. Exact field paths are in `../tables.json`. Intervals are marginal Clopper–Pearson intervals.

first reference-equivalent output or an invalid response. This stopping rule uses reference access and defines an evaluation procedure.

Success@3 means any successful post-feedback output, with every assigned problem retained in its arm denominator. It includes initially correct candidates. The planning MDE is approximately 15 percentage points at $n = 80$; it is not a treatment-effect confidence bound. Table 4 includes both strata and exact marginal intervals. The largest observed arm difference is 5 percentage points, below that planning threshold. Of 66 initially wrong A candidates, the arms repair 11, 12, and 14; of 14 initially correct candidates, 6, 6, and 5 have no successful post-feedback output. These are descriptive transitions.

Table 5 records compliance (membership of the preceding witness in the new feasible set) and damage (loss of a reference row previously implied by the nonempty candidate region). Witness acceptance and correct membership are separate: rejecting a reference-infeasible point can be correct. The verdict arm's witness is measured privately, not exposed. Undefined metrics retain null values and their reasons. First-round anchors reach 57 cases. The later rounds add zero successes; all valid third-round outputs equal their preceding candidates in the contract.

Metamorphic Alarms (T11)

The amended criterion evaluates alarms relative to published references, with no human false-positive estimate. Pre-outcome hash selection takes 300 distinct problems supporting all three transformations from 320 eligible triples. Units count groups of ten original units; annotated variable mentions become item A/B; explicit percentage/multiplier literals are rewritten using exact rational ratios. Unsupported expressions, including "times more," are excluded by rule. Unit scaling maps $x = 10z$; the other transformations preserve coordinates. Existing valid A outputs are reused as baselines. Source counts are 155 train, 32 development, and 113 test problems.

An alarm flags a transformed answer when it differs from the mapped baseline under the contract. A reference-conditional false alarm is an alarm whose transformed answer matches the mapped official model. Its denominator is all valid alarms: this is a false-discovery fraction, not the population false-positive rate. An initially wrong baseline followed by a correct transformed answer can cause it. Text meaning preservation is rule-based and was not independently human-validated.

Table 6 retains valid-pair denominators and invalid outputs. Pooled reference-conditional false alarms are 59/287 (20.56%), exceeding the preregistered 5% threshold; the

module is discarded. At least one violation occurs on 176/291 complete problems. Transformations of the same problem are correlated, so pooled alarms do not receive an independent-binomial interval.

Reproducibility

The facts sheet maps table cells and narrative macros to source fields. The public manuscript directory includes selected numerical outcomes, source hashes, and typesetting inputs. Raw prompts, responses, credentials, and source snapshots remain local under the repository's publication policy. Public checks reproduce derived aggregations; full geometric replay requires the archived raw inputs.

Running `python -m research.replay -check` from the project root reloads raw responses, replays frozen parsing and geometry, reconstructs the reading policies and selection baselines, regenerates both tables in memory, and compares saved structures and typeset cells exactly. It also checks that the frozen evidence matches the initial inventory. No inference service is contacted.

Related Work

NL4Opt concerns formulating optimization problems from natural-language descriptions (Ramamonjison et al. 2022). EquivaMap addresses equivalence through mappings between decision variables under Quasi-Karp equivalence (Zhai et al. 2025). Our fixed-variable contract addresses a narrower verification problem and does not evaluate variable-mapping discovery. The rational inclusion checks use standard linear-programming certificates; the empirical question here is how their feedback interacts with generated formulations and reference readings.

Limitations

- The empirical problems are 2-dimensional; support through dimension 6 has analytic regression coverage.
- The sensitivity policy uses 1 reviewer and outcome-selected cases; reference reuse adds 0 independent human labels.
- T8 uses 80 screened cases, fixed-peer feedback, reference-assisted stopping, and heuristic anchors; its observed gap is below the planning MDE.
- T11 uses 300 supported problems and rule-based text rewrites; its reference-conditional discard decision depends on 1 official source.
- Follow-ups use 1 model configuration and include public training problems; cross-model transfer and unseen-data generalization remain unmeasured.

T7 outcome	k/n	Rate (%)	Exact 95% CI (%)
All: modal success	76/100	76.00	66.43–83.98
Fully agreeing: error	13/83	15.66	8.61–25.29
Divergent: error	7/12	58.33	27.67–84.83
Incomplete: error	4/5	80.00	28.36–99.49

Table 3: Repeated sampling. All numerical cells derive from ../results.json, consistency.majority_success and consistency.conditional_errors. Incomplete cases remain in the overall denominator.

Stratum	Arm	Success/n	Rate (%)	Exact 95% CI (%)
All	verdict	19/80	23.75	14.95–34.58
All	point	20/80	25.00	15.99–35.94
All	point+anchor	23/80	28.75	19.18–39.95
Divergent	verdict	19/60	31.67	20.26–44.96
Divergent	point	20/60	33.33	21.69–46.69
Divergent	point+anchor	23/60	38.33	26.07–51.79
Agreeing but wrong	verdict	0/20	0.00	0.00–16.84
Agreeing but wrong	point	0/20	0.00	0.00–16.84
Agreeing but wrong	point+anchor	0/20	0.00	0.00–16.84

Table 4: T8 success within at most three rounds. Every numerical cell derives from ../results.json, ladder.main_table. Marginal intervals use distinct problems within each arm; they are not paired arm-difference intervals.

Arm	Round	Requests	Compliance k/n (CI%)	Damage k/n (CI%)
verdict	1	80	34/60 (43.2–69.4)	7/77 (3.7–17.8)
verdict	2	61	10/21 (25.7–70.2)	0/59 (0.0–6.1)
verdict	3	61	10/20 (27.2–72.8)	0/59 (0.0–6.1)
point	1	80	32/60 (40.0–66.3)	7/76 (3.8–18.1)
point	2	59	10/18 (30.8–78.5)	1/57 (0.0–9.4)
point	3	59	8/16 (24.7–75.3)	0/57 (0.0–6.3)
point+anchor	1	80	35/59 (45.7–71.9)	8/75 (4.7–19.9)
point+anchor	2	55	8/14 (28.9–82.3)	0/53 (0.0–6.7)
point+anchor	3	55	8/14 (28.9–82.3)	0/53 (0.0–6.7)

Table 5: T8 round measurements, with exact marginal intervals. Cells derive from ../results.json, ladder.main_table.all.*.rounds. Denominators exclude undefined metrics; stopped trajectories remain in the success denominator.

Transform	Violations/n	Rate (%)	Exact 95% CI (%)	False/alarms	Fraction (%)	Invalid
Unit scaling	139/294	47.28	41.45–53.16	18/139	12.95	6
Variable rename	64/297	21.55	17.01–26.67	20/64	31.25	3
Percentage/ratio	84/298	28.19	23.15–33.66	21/84	25.00	2

Table 6: T11 reference-conditional results. Cells derive from ../results.json, metamorphic.by_transformation. False alarms count reference-correct transformed answers among valid flags. Per-transform false-alarm intervals are also retained in the numerical source.

References

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